MASTER REPORT

# DHCP Failover Project Workflow

A condensed infrastructure report for Summit Care Medical Clinic focused on Windows Server 2022 DHCP, Cisco ASA emergency DHCP failover, Proxmox virtualization, DNS dependency planning, and recovery operations. The purpose of the project is to improve uptime, redundancy, and service continuity so the office can stay as operational as possible even if the primary DHCP server fails.

DELL POWEREDGE R740XD

PROXMOX TYPE 1 HYPERVISOR

WINDOWS SERVER 2022 DHCP

CISCO BACKUP DHCP

## Purpose & Goal

Build a supportable DHCP continuity design that reduces the chance of a single Windows service failure interrupting address delivery for office workstations, front-desk systems, and other wired clinic devices.

## Design Principle

Keep Windows Server as the primary DHCP authority, use Cisco only as an emergency path during outage conditions, and return cleanly to the normal service model after recovery.

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EXECUTIVE FOCUS

# What This Project Is Designed To Prove

This report presents Summit Care Medical Clinic as a realistic healthcare infrastructure scenario focused on one high-value objective: keep internal office devices receiving valid IP configuration even if the primary Windows DHCP service becomes unavailable, so the office can remain as operational as possible during outage conditions.

### Primary Service Model

Summit Care uses a Windows Server 2022 virtual machine on Proxmox as the normal DHCP and DNS authority for the internal office LAN. This keeps lease administration, DNS behavior, reservations, and support workflow centralized.

### Emergency Continuity Model

If the domain controller at 10.40.10.10 stops responding, the Cisco edge device enables an emergency DHCP pool so wired devices can still obtain an address, default gateway, and basic external connectivity while IT restores the primary server.

### Failback Discipline

When the Windows server comes back online and health checks stabilize, the Cisco backup pool is turned off again so the clinic returns to the primary service model and avoids long-term split administration.

> **Important design note:** this is not native Microsoft DHCP failover between two Windows servers. It is a layered continuity design that uses a Cisco backup service for outage coverage while preserving Windows as the primary DHCP authority.

OVERVIEW & STRATEGY

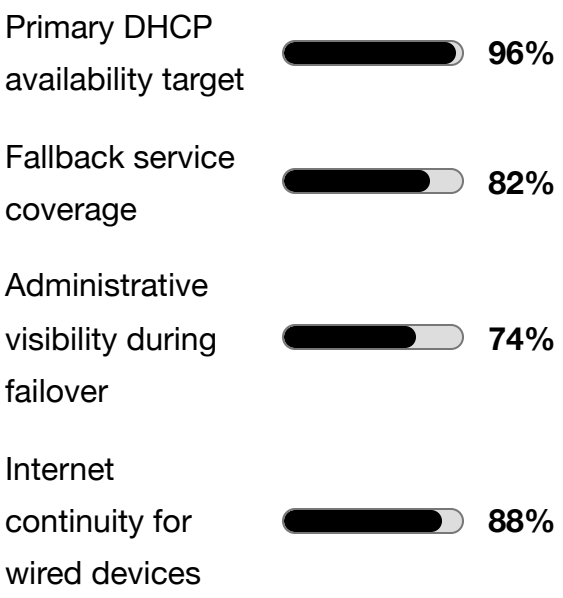
# Purpose, Goals & Summit Care Blueprint

The project goal is straightforward: improve uptime, reduce the operational impact of DHCP service failure, and keep the office infrastructure usable during an outage window. Rather than replacing Windows with Cisco, the strategy is to preserve Windows as the richer management platform while giving the clinic a practical emergency DHCP bridge on the network edge.

### Operational Objectives

- Reduce the chance that a single Windows DHCP outage stops new devices from joining the LAN.
- Support near-continuous address availability for front-desk, nursing, office, and wired workstation users.
- Preserve administrative clarity by automatically falling back to Windows DHCP after recovery.
- Support a more resilient office infrastructure posture built around uptime, redundancy, and cleaner failover recovery.

### Resilience Snapshot



### Summit Care Blueprint Snapshot

| Component | Role | Value To The Design |
|---|---|---|
| Dell PowerEdge R740xd | Physical compute host | Stable enterprise hardware base for critical infrastructure VMs |
| Proxmox VE | Type 1 hypervisor | Supports better recovery posture, snapshots, and host-level organization |
| Windows Server 2022 | Primary DHCP and DNS | Provides full administrative depth and domain-aware integration |
| Cisco edge firewall/router | Emergency DHCP backup | Supplies temporary lease continuity when the primary server is unavailable |

ARCHITECTURE & CONFIGURATION

## Platform Roles, Scope Design & Actual Build Workflow

The internal office LAN uses 10.40.10.0/24. Windows Server 2022 owns the normal lease range at 10.40.10.100 through 10.40.10.199, while Cisco is staged with a separate emergency-only range at 10.40.10.210 through 10.40.10.230. The default gateway remains 10.40.10.1, and the domain controller health-check IP remains 10.40.10.10.

| Setting | Example Value | Purpose |
|---|---|---|
| Network | 10.40.10.0/24 | Internal office LAN |
| Windows primary range | 10.40.10.100 - 10.40.10.199 | Main dynamic pool for desktops and office devices |
| Infrastructure exclusion | 10.40.10.1 - 10.40.10.99 | Preserves room for statics, printers, switches, and servers |
| Cisco emergency range | 10.40.10.210 - 10.40.10.230 | Emergency-only lease pool during outage conditions |
| Default gateway | 10.40.10.1 | Consistent traffic exit path in both normal and failover states |
| Primary DNS | 10.40.10.10 | Supports normal internal name resolution and domain-aware operation |

## Windows Server DHCP Manager Implementation

1. Build the Windows Server 2022 VM on Proxmox and assign 10.40.10.10 as a static address.
2. Install the DHCP Server role from Server Manager, complete post-install steps, and authorize the service in Active Directory.
3. Open DHCP Manager, create the IPv4 office scope for 10.40.10.0/24, and define the primary pool as 10.40.10.100 through 10.40.10.199.
4. Add exclusions for infrastructure space, then set router 10.40.10.1, DNS 10.40.10.10, lease timing, and any needed reservations.
5. Activate the scope and verify that a test client successfully receives a lease from Windows before introducing any backup logic.

## Cisco ASA Backup DHCP & Health-Based Activation

A realistic sample platform for this report is a Cisco ASA 5516-X or Firepower 1010 running ASA software. In this design, the ASA is not a full Windows-style failover partner. It is an emergency responder that is staged on the inside network and only activated when the Windows server path is unavailable.

| ASA Step | Configuration Intent | Why It Matters |
|---|---|---|
| Create backup pool | Stage 10.40.10.210 through 10.40.10.230 on the inside interface | Keeps emergency leases separate from the primary Windows range |
| Set DHCP options | Apply gateway, lease values, and outage-mode DNS behavior | Lets clients receive usable network settings during server loss |
| Keep inactive by default | Do not leave the ASA DHCP responder active during normal Windows service | Prevents split administration and overlapping lease behavior |
| Control with health checks | Only enable the backup pool when 10.40.10.10 is confirmed down | Makes failover conditional rather than permanent |
| Disable after recovery | Turn the emergency pool off when Windows is healthy again | Returns DHCP authority to the primary platform |

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sample ASA outage workflow

monitor 10.40.10.10 from the edge
if health checks fail beyond threshold:
  enable backup DHCP scope on inside interface

if health checks recover and remain stable:
  disable backup DHCP scope
  return lease authority to Windows Server
```

OPERATIONS & RECOVERY

# Trigger Logic, Testing Flow & Failback Routine

The health-check side of the workflow watches whether the domain controller at 10.40.10.10 is reachable. If reachability fails long enough to meet the outage threshold, the Cisco edge activates the emergency DHCP path so renewing or newly connected wired clients can still obtain valid IP configuration.

### Trigger Logic

1. Cisco repeatedly probes 10.40.10.10.
2. If the failure threshold is met, the emergency DHCP pool is enabled.
3. Clients renewing during the outage receive fallback leases from Cisco.
4. Once the server is healthy again, the backup pool is disabled and Windows resumes authority.

### Runbook Flow

1. Confirm 10.40.10.10 is unreachable from the Cisco edge.
2. Verify the emergency scope is active.
3. Test a wired client for lease issuance and outside internet access.
4. Restore the Windows VM or DHCP service and confirm stable response.
5. Verify Cisco disables the emergency scope and Windows retakes the primary role.

### Testing Checklist

- Simulate a DHCP service stop on Windows.
- Simulate a full Windows VM outage from the Proxmox side.
- Validate emergency lease delivery from the Cisco pool.
- Check whether wired clients keep basic internet reachability.
- Restore Windows and verify clean failback behavior.

> **Operational truth:** Cisco ASA DHCP is effective for emergency continuity, but it does not provide the same day-to-day administrative depth, reservation experience, or Windows-integrated DNS behavior as the primary server.

## RISK, GOVERNANCE & CONCLUSION

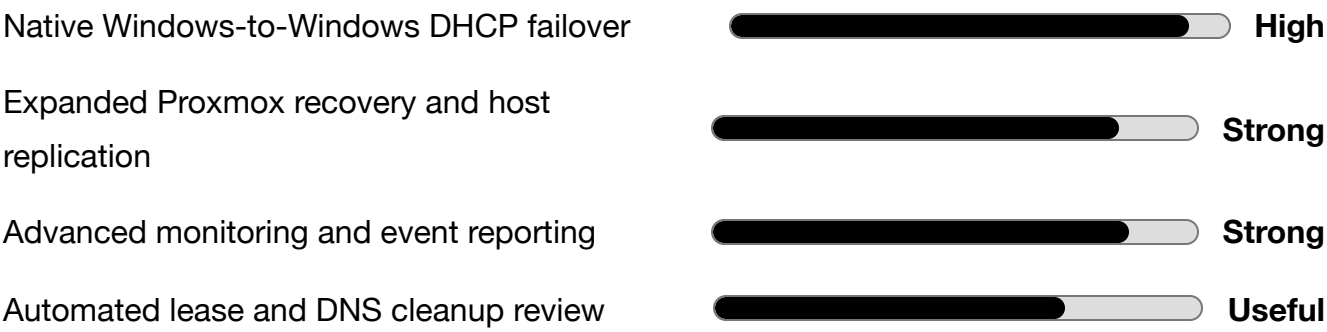
# Tradeoffs, Risk Controls & Final Engineering Takeaway

| Risk | Impact | Mitigation |
|---|---|---|
| Primary DHCP outage | New or rebooted devices may lose addressing | Cisco emergency pool keeps leases flowing during the outage |
| False failover activation | Confusion if both platforms appear active | Threshold-based health checks and separate address pools reduce the blast radius |
| Internal DNS degradation | Internet access may survive while internal names fail | Restore Windows quickly and document reduced outage-mode capability |
| Operational drift | Backup configuration becomes messy over time | Keep Cisco emergency-only and fail back to Windows automatically |

## What Cisco DHCP Does Less Well

- It offers less comfortable day-to-day lease visibility than DHCP Manager.
- It reduces Windows-integrated DNS, reservations, and policy depth during outage mode.
- It is appropriate as a continuity layer, not as the preferred long-term primary service.

## Improvement Roadmap



The final takeaway is that this design delivers a practical outage bridge for a healthcare office: Windows Server 2022 remains the preferred DHCP and DNS platform, Proxmox and UPS posture strengthen the primary path, and Cisco edge DHCP provides a temporary continuity layer when the primary server is down. It is honest about its limits, clear about its purpose, and organized as a professional infrastructure workflow.